

Session Info

Ivan Jacob Agaloos Pesigan

Session

```
sessionInfo()
```

```
#> R version 4.5.3 (2026-03-11)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p-r0.3.26.so; LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
#>  [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
#>  [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
#>  [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
#>  [9] LC_ADDRESS=C             LC_TELEPHONE=C
#> [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: Etc/UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  utils      datasets  methods    base
#>
#> other attached packages:
#> [1] rProject_0.0.25
#>
#> loaded via a namespace (and not attached):
#>  [1] backports_1.5.0  R6_2.6.1         lubridate_1.9.5  xfun_0.57
#>  [5] magrittr_2.0.4   glue_1.8.0       stringr_1.6.0    knitr_1.51
#>  [9] timechange_0.4.0 generics_0.1.4    lifecycle_1.0.5  xml2_1.5.2
#> [13] cli_3.6.5.9000  bibtex_0.5.2     compiler_4.5.3   plyr_1.8.9
#> [17] highr_0.12       rprojroot_2.1.1  httr_1.4.8       tools_4.5.3
#> [21] evaluate_1.0.5   Rcpp_1.1.1       RefManageR_1.4.0 otel_0.2.0
```

```
#> [25] rlang_1.1.7      jsonlite_2.0.0    stringi_1.8.7
```

Packages

```
unnname(installed.packages()[, 1])
```

```
#> [1] "simStateSpace"      "abind"              "Amelia"
#> [4] "arm"                "arrow"              "ash"
#> [7] "AsioHeaders"        "askpass"            "assertthat"
#> [10] "backports"          "base64enc"          "betaDelta"
#> [13] "betaMC"             "betaNB"             "betaSandwich"
#> [16] "BH"                 "bibtex"             "BiocManager"
#> [19] "BiocVersion"        "bit"                "bit64"
#> [22] "bitops"             "blob"               "bootStateSpace"
#> [25] "brew"              "brio"               "broom"
#> [28] "bslib"              "cachem"             "callr"
#> [31] "car"                "caracas"            "carData"
#> [34] "cellranger"         "cfrfr"              "checkmate"
#> [37] "cli"                "clipr"              "clock"
#> [40] "clusterGeneration" "coda"               "c0de"
#> [43] "collections"        "colorspace"         "commonmark"
#> [46] "conflicted"         "corpcor"            "covr"
#> [49] "cowplot"            "cpp11"              "crayon"
#> [52] "credentials"        "crosstalk"          "cTMed"
#> [55] "ctsem"              "curl"               "data.table"
#> [58] "DBI"                "dbplyr"             "Deriv"
#> [61] "desc"              "deSolve"            "devtools"
#> [64] "diagram"            "dials"              "DiceDesign"
#> [67] "diffobj"            "digest"             "distributional"
#> [70] "distrib"            "doBy"               "docopt"
#> [73] "downlit"            "dplyr"              "DT"
#> [76] "dtplyr"             "duckdb"             "dynr"
#> [79] "dynUtils"           "ellipse"            "ellipsis"
#> [82] "evaluate"           "expm"               "fans"
#> [85] "farver"             "fastDummies"        "fastmap"
#> [88] "fclust"             "fda"                "fdrtool"
#> [91] "fdr"                "fitVARMxID"         "FNN"
#> [94] "fontawesome"        "forcats"            "foreach"
#> [97] "forecast"           "Formula"            "fracdiff"
#> [100] "fs"                 "fst"                "fstcore"
#> [103] "furrr"              "future"             "future.apply"
#> [106] "gargle"             "generics"           "gert"
#> [109] "ggplot2"            "ggrepel"            "gh"
```

```

#> [112] "gitcreds"          "glasso"            "glmnet"
#> [115] "globals"          "glue"              "googledrive"
#> [118] "googlesheets4"    "gower"             "GPArotation"
#> [121] "GPfit"            "graphicalVAR"      "gridExtra"
#> [124] "gsubfn"           "gtable"            "gtools"
#> [127] "hardhat"          "haven"             "hdcrcde"
#> [130] "here"             "highr"             "Hmisc"
#> [133] "hms"              "htmlTable"         "htmltools"
#> [136] "htmlwidgets"      "httpgd"            "httpuv"
#> [139] "httr"             "httr2"             "ids"
#> [142] "ifaTools"         "igraph"            "infer"
#> [145] "ini"              "inline"            "ipred"
#> [148] "isoband"          "iterators"         "jomo"
#> [151] "jpeg"             "jquerylib"         "jsonlite"
#> [154] "jsonvalidate"     "kernlab"           "knitr"
#> [157] "ks"               "kutils"            "labeling"
#> [160] "Lahman"           "languageserver"    "later"
#> [163] "latex2exp"        "lava"              "lavaan"
#> [166] "lazyeval"         "lhs"               "lifecycle"
#> [169] "lintr"            "lisrelToR"         "listenv"
#> [172] "litedown"         "littler"           "lme4"
#> [175] "lmtest"           "locfit"            "longMI"
#> [178] "loo"              "lubridate"         "magick"
#> [181] "magrittr"         "markdown"          "MatrixModels"
#> [184] "matrixStats"      "mclust"            "memoise"
#> [187] "metaDyn"          "metaSEM"           "mi"
#> [190] "mice"             "microbenchmark"    "mime"
#> [193] "miniUI"           "minqa"             "mitml"
#> [196] "mize"             "mlVAR"             "mnormt"
#> [199] "modeldata"        "modelenv"          "modelr"
#> [202] "MplusAutomation" "multicool"         "mvtnorm"
#> [205] "nloptr"           "numDeriv"          "nycflights13"
#> [208] "OpenMx"           "openssl"           "openxlsx"
#> [211] "ordinal"          "otel"              "pak"
#> [214] "pan"              "pander"            "parallelly"
#> [217] "parsnip"          "patchwork"         "pbapply"
#> [220] "pbivnorm"         "pbkrtest"          "pcaPP"
#> [223] "pdftools"         "pillar"            "pkgbuild"
#> [226] "pkgconfig"        "pkgdown"           "pkgload"
#> [229] "plyr"             "png"               "posterior"
#> [232] "pracma"           "praise"            "prettyunits"
#> [235] "printr"           "processx"          "prodlim"
#> [238] "profvis"          "progress"          "progressr"
#> [241] "promises"         "proto"             "ps"
#> [244] "psych"            "purrr"             "qgraph"

```

```

#> [247] "qpdf" "quadprog" "quantreg"
#> [250] "quarto" "QuickJSR" "R.cache"
#> [253] "R.methodsS3" "R.oo" "R.utils"
#> [256] "R2jags" "R2WinBUGS" "R6"
#> [259] "ragg" "rainbow" "rappdirs"
#> [262] "rbibutils" "rcmdcheck" "RColorBrewer"
#> [265] "Rcpp" "RcppArmadillo" "RcppEigen"
#> [268] "RcppGSL" "RcppParallel" "RcppTOML"
#> [271] "RCurl" "Rdpack" "readr"
#> [274] "readxl" "recipes" "RefManageR"
#> [277] "reformulas" "rematch" "rematch2"
#> [280] "remotes" "reprex" "reshape2"
#> [283] "reticulate" "rex" "rhdf5"
#> [286] "rhdf5filters" "Rhdf5lib" "rhub"
#> [289] "rjags" "rlang" "RMariaDB"
#> [292] "rmarkdown" "rockchalk" "roxygen2"
#> [295] "rpf" "RPostgres" "rProject"
#> [298] "rprojroot" "rsample" "RSQLite"
#> [301] "rstan" "rstantools" "rstudioapi"
#> [304] "RUnit" "rversions" "rvest"
#> [307] "Ryacas" "S7" "sass"
#> [310] "scales" "selectr" "sem"
#> [313] "semlbci" "semmcci" "semPlot"
#> [316] "sessioninfo" "sfd" "shape"
#> [319] "shiny" "simStateSpace" "slider"
#> [322] "snow" "snowfall" "sourcetools"
#> [325] "SparseM" "sparsevctrs" "SQUAREM"
#> [328] "StanHeaders" "stringi" "stringr"
#> [331] "styler" "symSEM" "sys"
#> [334] "systemfonts" "tailor" "tensorA"
#> [337] "testthat" "texreg" "textshaping"
#> [340] "tibble" "tidymodels" "tidyr"
#> [343] "tidyselect" "tidyverse" "timechange"
#> [346] "timeDate" "tinytex" "tune"
#> [349] "tzdb" "ucminf" "unigd"
#> [352] "urca" "urlchecker" "usethis"
#> [355] "utf8" "uuid" "V8"
#> [358] "vctrs" "viridisLite" "vroom"
#> [361] "waldo" "warp" "whisker"
#> [364] "whoami" "withr" "workflows"
#> [367] "workflowsets" "xfun" "XML"
#> [370] "xml2" "xmlparsedata" "xopen"
#> [373] "xtable" "yaml" "yardstick"
#> [376] "zip" "zoo" "base"
#> [379] "boot" "class" "cluster"

```

```

#> [382] "codetools"      "compiler"      "datasets"
#> [385] "foreign"        "graphics"      "grDevices"
#> [388] "grid"          "KernSmooth"   "lattice"
#> [391] "MASS"          "Matrix"        "methods"
#> [394] "mgcv"          "nlme"          "nnet"
#> [397] "parallel"      "rpart"         "spatial"
#> [400] "splines"       "stats"         "stats4"
#> [403] "survival"      "tcltk"         "tools"
#> [406] "utils"

```

References

- Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. <https://doi.org/10.1037/met0000779>
- R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. <https://www.R-project.org/>